A Structural Causal Model for Goal-Frontier Maximization

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Abstract

The goal-frontier maximization (GFM) sequence establishes a computable alignment objective with structural safety properties and endogenous anti-monopolar pressure. The preceding papers treat agent evaluation as a statistical signal-processing problem: trust-weighted votes, correlational contraction detection, max-aggregated risk reports. This paper introduces a structural causal model for capability dynamics and uses it to sharpen every agent-evaluation mechanism, upgrading the core identification results from statistical to causal under explicitly stated structural assumptions. Do-calculus contraction attribution sharpens scorpion detection under additive-SCM, causal-sufficiency, and exogenous-orthogonality conditions. Pathway-conditional residuals formalize risk-trust with L^2 convergence guarantees. A log-linear opinion pool replaces max-aggregation without discarding independence information. A probabilistic dependency-risk model sharpens the minimax substrate bound into a per-step expected cost that compounds with growth, gated by an adversarial balance condition. Each upgrade follows the same pattern: a statistical proxy becomes a counterfactual query on the actor's own causal model.

1 Introduction

Goal-frontier maximization (GFM) proposes alignment through a single objective: maximize the volume of the jointly achievable capability space $\mathrm{vol}_{\mathrm{P}}(G)$ [Lasser, 2026a]. The foundational paper proves structural safety properties (self-balancing against destruction, coercion, and rigidity). Its companion [Lasser, 2026c] constructs a computable capability poset with polynomial-time measure computation and a leverage ranking signal that produces a gradient toward civilization-building infrastructure. The third paper [Lasser, 2026b] extends the framework to multi-step planning through a discounted value function, develops risk-capability identification, and establishes the anti-monopolar property: under a diversity condition, full capability domination is anti-maximizing.

The preceding papers treat every agent-evaluation question as a statistical signal-processing problem. The foundational paper's scorpion detection [Lasser, 2026a] identifies agents whose actions are *correlated* with $\mathrm{vol}_{\mathrm{P}}$-contraction but cannot disentangle their contribution from simultaneous causes. The horizon paper's risk-trust factor ${T^{\mathrm{risk}}}_j$ [Lasser, 2026b] has no formal update dynamics. The dependency-risk analysis uses minimax bounds that are quantitatively too weak for risk-adjusted planning. In each case, statistical signals suffice for obvious cases (direct harm, clear deception) but fail for the hard cases that matter most: confounded contraction in multi-agent populations, coordinated risk manipulation, and risk claims whose outcomes are never observed.

This paper introduces a structural causal model for capability dynamics and uses it to sharpen every agent-evaluation mechanism, upgrading the core identification results from statistical to causal. The common thread: a GFM agent needs a *causal model* of other agents to make good decisions about their harm (who caused this contraction?), credibility (is this risk claim structurally sound?), and adversarial exposure (what substrate-correlated threats do we face?).

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Contributions.

1. A structural causal model for capability dynamics with do-calculus contraction attribution, showing a conditional L^2 convergence advantage over correlational detection under causal sufficiency, additive-SCM structure, and exogenous orthogonality (Section 2).
2. L^2 EWMA dynamics for risk-trust ${T^{\text{risk}}}_j$ over pathway-conditional structural verification residuals, and a prior-corrected trust-weighted log-linear opinion pool that recovers Bayesian updating in the no-tempering limit, replacing the max aggregation operator, with a two-gate architecture separating attention allocation from action decisions via the actor's own forward causal risk evaluation (Section 3).
3. Causal scorpion detection under a Gaussian model, committed to a sequential probability ratio test on least-favorable simple hypotheses with Type I/II error control, and a high-probability bound on non-stationary scorpion evasion bandwidth under a stationary-epoch model (Section 4).
4. A probabilistic dependency-risk model gated by an adversarial balance condition $c > \max_i(f_i \cdot c_i)$ that admits contagion dynamics, with per-step expected costs that compound at the same rate as growth and a quantitative substrate-count lower bound $m > c_{\max}/c$ that sharpens the minimax $m \geq 2$ (Section 5).

2 Causal Attribution Framework

The foundational paper's scorpion detection (Proposition 2 of Lasser [2026a]) identifies agents whose actions are *correlated* with $\mathrm{vol}_{\mathrm{P}}$-contraction. It explicitly acknowledges that "disentangling the agent's contribution from simultaneous causes requires a causal-attribution procedure that the framework does not provide." This section provides that procedure.

2.1 Structural Causal Model for Capability Dynamics

Definition 1 (Capability Dynamics SCM)**.** *A* structural causal model for capability dynamics *is a tuple $\mathcal{M} = (U, V, F, P(U))$ where:*

- *U: exogenous variables representing agent dispositions, environmental conditions, and stochastic factors not determined by the model.*
- *$V = \{G_t, \pi_t^{(1)}, \ldots, \pi_t^{(n)}, \Delta\mathrm{vol}_{\mathrm{P}t}\}$: endogenous variables representing the capability poset state, each agent's action at time t, and the resulting $\mathrm{vol}_{\mathrm{P}}$-change.*
- *F: structural equations mapping parent variables to children in the causal directed acyclic graph (DAG). The DAG encodes which agents' actions causally influence which capability changes.*
- *$P(U)$: prior distribution over exogenous variables.*

When the variance-reduction results of this paper are invoked (Proposition 1(c) and downstream), we additionally assume ***exogenous orthogonality****: the exogenous noise components entering different agents' structural equations are mutually independent conditional on G_t. That is, if $U_j \subseteq U$ are the exogenous variables appearing in a_j's structural equation and $U_k \subseteq U$ those appearing in a_k's, then $U_j \perp\!\!\!\perp U_k \mid G_t$ for $j \neq k$. This rules out shared exogenous shocks that would induce covariance between agents' contributions even under a correct causal DAG.*

The GFM actor's world model $\mathcal{W}$ (Definition 1 of Lasser [2026b]) is an instance of $\mathcal{M}$ restricted to the actor's beliefs about causal structure.

The SCM extends $\mathcal{W}$ by making the causal structure explicit. In the foundational paper, $\mathcal{W}$ is an opaque prediction model: given actions, it predicts $\mathrm{vol}_{\mathrm{P}}$-changes. The SCM decomposes predictions into causal pathways, enabling counterfactual queries: "what would have happened if agent a_j had acted differently?"

Temporal scope. Definition 1 specifies a *single-step* SCM: the endogenous variables $\{G_t, \pi_t^{(1)}, \ldots, \pi_t^{(n)}, \Delta\mathrm{vol}_{\mathrm{P}t}\}$ describe one time slice, and the structural equations F map actions at step t to the $\mathrm{vol}_{\mathrm{P}}$-change at step t. The do-intervention $\mathrm{do}(\pi_j = \cdot)$ operates within this single-step structure: "other variables respond according to F" means other endogenous variables in the same time slice are recomputed under the intervened structural equations, not that the system evolves forward in time. Multi-step effects (an action at t that causes a cascade at $t+1$) are captured by instantiating a new single-step SCM at each subsequent time step with an updated poset state G_{t+1}; the causal attribution $\mathrm{CA}(a_j, t)$ (Definition 2) is the single-step effect.

2.2 Causal Contraction Attribution

Definition 2 (Causal Contraction Attribution)**.** *For an observed* $\mathrm{vol}_{\mathrm{P}}$*-change* $\Delta\mathrm{vol}_{\mathrm{P}}(G_t)$ *at time t, the* causal contribution *of agent* a_j *is:*

$$\mathrm{CA}(a_j, t) = \Delta\mathrm{vol}_{\mathrm{P}}\big(G_t \mid \mathrm{do}(\pi_j = \pi_j^{\mathrm{obs}})\big) - \Delta\mathrm{vol}_{\mathrm{P}}\big(G_t \mid \mathrm{do}(\pi_j = \mathrm{skip})\big) \tag{1}$$

where $\mathrm{do}(\pi_j = \cdot)$ *denotes the do-calculus intervention [Pearl, 2009]: setting* a_j*'s action to a specific value while allowing all other variables to respond according to* F*. The first term is the* $\mathrm{vol}_{\mathrm{P}}$*-change under* a_j*'s observed action; the second is the counterfactual* $\mathrm{vol}_{\mathrm{P}}$*-change had* a_j *done nothing.*

The causal attribution $\mathrm{CA}(a_j, t)$ isolates a_j's contribution from confounders: simultaneous actions by other agents, environmental changes, and background dynamics that the correlational signal in Proposition 2 of Lasser [2026a] cannot separate. When a_j's actions cause contraction, $\mathrm{CA}(a_j, t) < 0$. When they cause expansion, $\mathrm{CA}(a_j, t) > 0$. When a_j's actions have no causal effect (the contraction would have occurred regardless), $\mathrm{CA}(a_j, t) = 0$, and the correlational detection would have produced a false positive.

Additivity. The one-at-a-time $\mathrm{CA}(a_j, t)$ of Equation (1) sums to the total $\mathrm{vol}_{\mathrm{P}}$-change only when the SCM's structural equation for $\Delta\mathrm{vol}_{\mathrm{P}}$ is *additive in agent actions*: there exist functions g_j and a residual h such that

$$\Delta\mathrm{vol}_{\mathrm{P}}(G_t) = \sum_j g_j\big(\pi_j^{\mathrm{obs}}, G_t, U\big) + h(G_t, U) \tag{2}$$

where h does not depend on any π_j. Under this additive-SCM assumption, the one-at-a-time intervention $\mathrm{do}(\pi_j = \pi_j^{\mathrm{obs}})$ vs. $\mathrm{do}(\pi_j = \mathrm{skip})$ isolates g_j exactly, and

$$\Delta\mathrm{vol}_{\mathrm{P}}(G_t) = \sum_j \mathrm{CA}(a_j, t) + h(G_t, U). \tag{3}$$

The additive-SCM assumption is stronger than (and is not implied by) conditional independence of agent actions given G_t: two independently drawn actions can still interact multiplicatively in a non-additive structural equation such as $\Delta\mathrm{vol}_{\mathrm{P}} = \pi_1 \cdot \pi_2$, in which case the one-at-a-time CA does not sum to the total. When interaction effects are present and a non-additive SCM is required, an exact decomposition needs a Shapley-style attribution rule that averages CA over all orderings of agent inclusion [Shapley, 1953]; the one-at-a-time form is then an approximation whose error is the deviation from additivity. For scorpion detection (Section 4), the one-at-a-time form is sufficient because the detection signal depends only on the sign and magnitude of the diagonal contribution g_j, not on the exact decomposition of interaction effects.

2.3 Causal Detection Dominates Correlational Detection

Proposition 1 (Convergence Advantage of Causal Attribution)**.** *Let* a_j *be an agent with expected causal contraction* $\mu_j = \mathbb{E}[\mathrm{CA}(a_j, t)] < 0$ *(a scorpion). Assume the observation sequences for both detectors are i.i.d. within a stationary strategy epoch, or weakly dependent with summable autocovariance. Under the SCM* $\mathcal{M}$*, the following* information-theoretic *sample-complexity comparison holds (independent of the specific sequential test used; see Section 4 for the SPRT-specific operational rates):*

- ***(a)*** *The correlational detector (Proposition 2(a) of Lasser [2026a]) has sample complexity* $\Omega(\sigma^2/\mu_j^2)$ *for identifying* a_j *as a net-contractor, under the idealized assumption that* μ_j

appears as an isolable drift in the trust-weighted aggregate $\hat{\Delta}$ (see paragraph below on the correlational estimator). Here σ^2 is the total variance of the signal the correlational detector can construct, including confounders.

(b) *The causal detector using $\mathrm{CA}(a_j, t)$ has sample complexity $\Omega(\sigma^2_{\mathrm{do}}/\mu_j^2)$, where σ^2_{do} is the variance of the causal attribution signal after conditioning on the do-intervention. The ratio $\sigma^2_{\mathrm{do}}/\sigma^2 \leq 1$ (under the conditions of part (c)) is the structural advantage factor.*

(c) *Under **causal sufficiency** (the SCM $\mathcal{M}$ captures all common causes of a_j's actions and the confounder signals), the **additive-SCM** assumption (Equation (2)), and **exogenous orthogonality** (Definition 1), we have $\mathrm{Cov}(\mathrm{CA}(a_j), \mathrm{confounders} \mid G_t) = 0$ and therefore*

$$\sigma^2 - \sigma^2_{\mathrm{do}} = \mathrm{Var}(\textit{confounders} \mid G_t) \geq 0,$$

so $\sigma^2_{\mathrm{do}} \leq \sigma^2$, with strict inequality whenever confounders contribute nonzero variance. Without causal sufficiency (or without the additive-SCM structure), the variance identity gives

$$\sigma^2 - \sigma^2_{\mathrm{do}} = \mathrm{Var}(\textit{confounders} \mid G_t) + 2\,\mathrm{Cov}(\mathrm{CA}(a_j), \mathrm{confounders} \mid G_t),$$

whose sign depends on the covariance term. The inequality $\sigma^2_{\mathrm{do}} \leq \sigma^2$ therefore requires the full triple (causal sufficiency, additive SCM, exogenous orthogonality), or the weaker condition $\mathrm{Cov}(\mathrm{CA}(a_j), \mathrm{confounders} \mid G_t) \geq -\frac{1}{2}\mathrm{Var}(\textit{confounders} \mid G_t)$. Under a general (non-sufficient) SCM with adversarial confounder correlation, causal attribution can in principle achieve worse convergence than the correlational detector; characterizing populations in which this occurs is outside the scope of this paper.

Proof. (a) The correlational detector's observable signal is the trust-weighted aggregate $\hat{\Delta}(\pi_t) = \sum_{k \in \mathcal{O}} T_k \cdot R_k(\pi_t)$ (Equation 10 of Lasser [2026a]), where $R_k \in \{-1, 0, +1\}$ is a ternary directional vote. Let $\Delta\mathrm{vol}_{\mathrm{P}\,\mathrm{total}}(t)$ denote the *true* continuous $\mathrm{vol}_{\mathrm{P}}$-change at step t that $\hat{\Delta}$ estimates. The correlational detector's sample complexity for detecting a_j's mean effect μ_j is governed by the variance of the continuous quantity it must recover, not by the variance of the ternary estimator itself: a sign-correct ternary estimator inherits the variance of the underlying continuous signal it tracks, plus discretization noise. Formally, σ^2 in the bound is the variance of any estimator of the total $\mathrm{vol}_{\mathrm{P}}$-change that the correlational detector can construct from its observation channel. The $\Omega(\sigma^2/\mu_j^2)$ lower bound is the information-theoretic sample complexity for detecting a mean shift μ_j in noise of variance σ^2.

(b) The causal detector evaluates $\mathrm{CA}(a_j, t)$ directly by comparing the observed $\mathrm{vol}_{\mathrm{P}}$-change against the counterfactual under $\mathrm{do}(\pi_j = \mathrm{skip})$. The variance of this signal is $\sigma^2_{\mathrm{do}} = \mathrm{Var}(\mathrm{CA}(a_j, t))$, which excludes the confounding variance.

(c) Under the additive-SCM assumption (Equation (2)), the continuous $\mathrm{vol}_{\mathrm{P}}$-change decomposes as $\Delta\mathrm{vol}_{\mathrm{P}\,\mathrm{total}}(t) = \mathrm{CA}(a_j, t) + \mathrm{confounders}(t)$, where the confounder term aggregates all other agents' contributions and the residual $h(G_t, U)$. By the variance identity for sums:

$$\sigma^2 = \mathrm{Var}(\mathrm{CA}(a_j)) + \mathrm{Var}(\mathrm{confounders}) + 2\,\mathrm{Cov}(\mathrm{CA}(a_j), \mathrm{confounders}).$$

Under causal sufficiency, the additive-SCM structure, and exogenous orthogonality (Definition 1): causal sufficiency ensures the DAG captures all common causes, the additive decomposition ensures a_j's contribution enters as a separable term g_j, and exogenous orthogonality ensures the noise driving g_j is independent of the noise driving the confounder terms $\{g_k\}_{k \neq j}$ and h. Together, these give $\mathrm{Cov}(\mathrm{CA}(a_j), \mathrm{confounders}) = 0$, and $\sigma^2 - \sigma^2_{\mathrm{do}} = \mathrm{Var}(\mathrm{confounders})$ as stated. Without causal sufficiency, the covariance term is retained and the inequality $\sigma^2_{\mathrm{do}} \leq \sigma^2$ holds only when $\mathrm{Var}(\mathrm{confounders}) + 2\,\mathrm{Cov}(\mathrm{CA}(a_j), \mathrm{confounders}) \geq 0$; the inequality is strict whenever confounders have positive variance and the covariance term does not dominate. ■

The correlational detector as an estimator. Part (a) of Proposition 1 treats the correlational detector as if μ_j appeared directly as a component of the mean of the aggregate signal $\hat{\Delta}$. This is an idealization: the actual mean of $\hat{\Delta}$ is the sum of all agents' mean effects plus background dynamics, not μ_j alone. A correlational estimator that recovered μ_j from $\hat{\Delta}$ would need an additional auxiliary

procedure — for example, regressing $\hat{\Delta}$ against an indicator for a_j's action times, which incurs its own bias when confounders are correlated with a_j's action schedule. The $O(\sigma^2/\mu_j^2)$ bound therefore describes the best-case correlational detector: one that has access to a_j's action schedule and uses it as a conditioning variable. A weaker correlational detector that only observes $\hat{\Delta}$ without conditioning has a strictly worse (and generally incomputable) rate. The causal detector does not need this auxiliary procedure because $\mathrm{CA}(a_j, t)$ already isolates a_j's contribution by construction.

Practical implication. In populations with many simultaneously acting agents, confounding variance can dominate the correlational signal. A single scorpion among 100 simultaneously acting agents contributes $\sim$1% of the signal variance; the correlational detector needs $\sim$100$\times$ more observations than the causal detector to achieve the same confidence. The causal detector's advantage grows linearly with the number of simultaneously acting confounders.

2.4 Causal Attribution as One Channel Among Several

Proposition 1 assumes that the SCM $\mathcal{M}$ is accurate enough to support do-calculus attribution, and that the structural assumptions of part (c) (causal sufficiency and additive-SCM structure) hold. In practice the SCM is learned from observational data, partially specified, and adversarially exposed: agents can strategically align their actions to mask causal structure, and new agents or delegation chains can emerge during the trajectory. A framework that depended on the SCM being perfectly learned would be brittle. Instead, the framework is robust because causal attribution is *one channel* among several in a multi-channel attribution system, and the system's overall attribution accuracy degrades gracefully as any single channel's reliability degrades.

Multi-channel attribution is the standard methodology in adversarial intrusion analysis and intelligence tradecraft. The Diamond Model of intrusion analysis [Caltagirone et al., 2013] organizes attribution around four analytic vertices — adversary, capability, infrastructure, victim — linked by relationships, with confidence coming from cross-vertex correlation rather than from any single vertex. The *Pyramid of Pain* [Bianco, 2013] ranks observable signals by how costly they are for an adversary to change: hash values, IP addresses, and domain names are cheap to rotate; tools and tactics, techniques, and procedures (TTPs) are expensive to retrain. The MITRE ATT&CK framework [Strom et al., 2018] provides a shared capability taxonomy that grounds TTP-level attribution in a common vocabulary. The *Analysis of Competing Hypotheses* [Heuer, 1999] provides a structured method for comparing attribution hypotheses by diagnostic evidence, weighting signals by their power to discriminate among candidates rather than by their individual strength. In all four traditions, the robustness of an attribution decision is driven by signals the adversary cannot cheaply fake, and no single signal is load-bearing.

The GFM sequence already supplies several attribution channels, distributed across papers:

- **Behavioral prediction residual.** The trust model of Lasser [2026a] computes $r_k(t) = R_k(t) - \hat{R}_k(t \mid M_k(G))$, the divergence between an agent's observed behavior and its predictive model. An agent whose actions consistently match its model concentrates on high T_k; systematic divergence drives T_k down. This is a moderately expensive channel to fake: a scorpion must maintain behavioral consistency with a fictitious model to evade it.
- **Correlational co-occurrence.** The trust-weighted $\hat{\Delta}$ signal of Lasser [2026a] (Proposition 2(a)) detects agents whose actions temporally co-occur with $\mathrm{vol_P}$-contraction. This is a cheap channel to fake — an adversary can introduce uncorrelated dummy actions — and is strictly dominated by the causal channel under the conditions of Proposition 1(c).
- **Capability gating.** Papers 2 and 3 of the GFM sequence [Lasser, 2026c,b] together provide a structural constraint: an agent cannot cause the contraction of a capability it does not have upstream access to, regardless of any other signal. This is the *means-analysis* from the intrusion-analysis tradition and sits at the top of the robustness hierarchy: it reflects the structure of the poset itself rather than behavior an adversary can stage.[1] An attribution decision that conflicts with capability gating should be overridden by it, not aggregated with it.

[1] The gate's strength is bounded by the fidelity of the maintained capability model: hidden delegation, latent substitute capabilities, or a misspecified poset can defeat it.

- **Causal contraction attribution** (this paper, Definition 2). Moderately expensive to fake, since the SCM demands counterfactual consistency, and strictly informative under the conditions of Proposition 1(c). Vulnerable to SCM misspecification.
- **SPRT scorpion detection** (this paper, Section 4). Moderately expensive to fake under stationary epochs; vulnerable to fast adaptation as characterized by the evasion bandwidth bound.

Table 1 summarizes the channels by evasion cost and required assumptions.

Channel	Cost to fake	Paper	Key assumptions
Capability gating	Very high	2, 3	Trusted, complete capability graph
Causal attribution	Moderate	4	Causal sufficiency, additive SCM, exog. orthogonality
Behavioral residual	Moderate	1	Predictive-model access
SPRT detection	Moderate	4	Gaussian, stationary epochs
Correlational co-occ.	Low	1	(None beyond observation)

Table 1: Attribution channels ordered by adversarial evasion cost (highest first), following the Pyramid of Pain principle.

An attribution decision should aggregate evidence across channels, weighted by each channel's calibrated reliability, rather than rely on the output of any single channel. (The aggregation is Bayesian in spirit — each channel contributes evidence toward or against a suspect — but this paper does not formalize the joint hypothesis space or channel-dependence model; the full fusion rule is an open direction identified below.) The multi-channel framing has three architectural consequences:

1. **Capability gating is a hard filter, not a prior.** An agent that cannot structurally reach capability c_i cannot have caused its contraction, and no amount of causal or correlational evidence should override that. The correct aggregation is a gate-then-aggregate architecture: apply the capability gate first to produce the candidate suspect set, then aggregate the remaining channels over the candidates.
2. **No single channel must be perfect.** Each channel has its own reliability, and the aggregate's accuracy depends on the joint reliability profile rather than on the maximum-reliability channel. This is the property that makes multi-channel attribution robust in the threat-intelligence setting and that makes the GFM actor robust to imperfect structure learning (Section 6).
3. **Channel weights should be calibrated from measurable proxies**, not fixed at design time. The causal channel's weight should scale with an SCM-confidence estimate derived from structure-learning diagnostics; the behavioral channel's weight should scale with a population-level reliability measure derived from the per-agent trust scores T_k of Lasser [2026a] (distinct from the actor's self-trust T_s, which governs learning rate rather than channel reliability); the correlational channel's weight should scale with the confounding variance estimated from the population structure. The calibration procedure is outside the scope of this paper and is identified as an open direction in Section 6.

Paper 4's contribution to this framework is the causal channel: a principled, do-calculus-based attribution signal with a convergence advantage over the correlational baseline under stated structural conditions. It is not the attribution system, and the paper does not claim that the causal channel is load-bearing. The overall attribution robustness comes from the multi-channel structure, and from capability gating in particular as the structural floor.

3 Risk-Trust Dynamics

The companion paper [Lasser, 2026b] introduces a risk-trust factor ${T^{\text{risk}}}_j$ for weighting communicated risk claims but provides only a qualitative update rule. This section formalizes the dynamics, drawing on the EWMA structure of the foundational paper's behavioral trust [Lasser, 2026a].

3.1 Structural Verification Residuals

Definition 3 (Structural Verification Residual)**.** *For a risk claim $(S, \mathcal{P}, |\Delta\mathrm{vol}_{\mathrm{P}}|, p)$ from agent a_j (Definition 4 of Lasser [2026b]), the actor evaluates* the specific claimed pathway *$\mathcal{P}$ against its own SCM $\mathcal{M}$ (Definition 1) using the causal risk evaluation procedure (Section 4.3 of Lasser [2026b]). This produces a pathway-conditional assessment $\big(p_{\mathcal{W}}(\mathcal{P}),\ |\Delta\mathrm{vol}_{\mathrm{P}\,\mathcal{W}}(\mathcal{P})|\big)$: the probability and contraction magnitude the actor would assign if $\mathcal{P}$ were the realized causal pathway. If $\mathcal{P}$ has no counterpart in $\mathcal{M}$ — that is, if the actor's causal model contains no edges implementing the claimed mechanism — then $\big(p_{\mathcal{W}}(\mathcal{P}), |\Delta\mathrm{vol}_{\mathrm{P}\,\mathcal{W}}(\mathcal{P})|\big) = (0, 0)$ by convention. The* structural verification residual *is the dimensionless two-component divergence:*

$$r_j^{\mathrm{risk}}(t) = \left(p_j^{\mathrm{claimed}} - p_{\mathcal{W}}(\mathcal{P}),\ \ \frac{|\Delta\mathrm{vol}_{\mathrm{P}}{}_j^{\mathrm{claimed}}| - |\Delta\mathrm{vol}_{\mathrm{P}\,\mathcal{W}}(\mathcal{P})|}{\mathrm{vol}_{\mathrm{P}}(G_t)} \right) \tag{4}$$

Each component is in $[-1, 1]$: the first is a probability difference, the second is a relative volume difference normalized by the current total poset volume. The Euclidean norm $\|r_j^{\mathrm{risk}}(t)\|$ is therefore dimensionally consistent and invariant under rescaling of $\mathrm{vol}_{\mathrm{P}}$. A fabricated pathway that does not exist in $\mathcal{M}$ produces the maximal residual $\|r_j^{\mathrm{risk}}(t)\| = \sqrt{(p_j^{\mathrm{claimed}})^2 + (|\Delta\mathrm{vol}_{\mathrm{P}}{}_j^{\mathrm{claimed}}|/\mathrm{vol}_{\mathrm{P}}(G_t))^2}$; structural verification is enforced by the pathway-conditional evaluation, not by an extra pathway-comparison term.

Unlike the behavioral prediction residual $r_k(t)$ (Equation 25 of Lasser [2026a]), which compares *predicted* against *observed* behavior, the risk residual compares *claimed* against *independently modeled* risk. This is necessary because risk claims are counterfactual predictions that may never be directly observed — the *Wamura problem*, introduced in the poset paper [Lasser, 2026c] and developed in the risk-communication analysis of Lasser [2026b, §4]: a successful risk intervention removes the very evidence that would have justified it, so the convergence timescale for risk-trust can in principle exceed the timescale on which trust normally accumulates. The SCM provides the independent assessment needed to break this circularity: the actor's own causal model produces $(p_{\mathcal{W}}, |\Delta\mathrm{vol}_{\mathrm{P}\,\mathcal{W}}|)$ without relying on the claim.

3.2 EWMA Risk-Trust Update

Definition 4 (Risk-Trust Dynamics)**.** *The risk-trust factor $T^{\mathrm{risk}}{}_j$ evolves through an exponentially weighted moving average of squared structural verification residuals:*

$$\sigma_j^{\mathrm{risk},2}(t) = \alpha_{\mathrm{risk}} \cdot \sigma_j^{\mathrm{risk},2}(t-1) + (1 - \alpha_{\mathrm{risk}}) \cdot \|r_j^{\mathrm{risk}}(t)\|^2 \tag{5}$$

$$T^{\mathrm{risk}}{}_j(t) = \frac{1}{1 + \beta_{\mathrm{risk}} \cdot \sigma_j^{\mathrm{risk},2}(t)} \tag{6}$$

where $\alpha_{\mathrm{risk}} \in (0, 1)$ controls the decay rate (tracking speed vs. stability), $\beta_{\mathrm{risk}} > 0$ controls the sensitivity (how rapidly trust degrades with inconsistency), and $\|\cdot\|$ is the Euclidean norm on the two-dimensional residual.

The structure mirrors the foundational paper's behavioral trust exactly (Equations 33 and 38 of Lasser [2026a]), with the behavioral prediction residual replaced by the structural verification residual. The parameters α_{risk} and β_{risk} are independent of the behavioral parameters α and β: an agent can have high behavioral trust T_k (predictable behavior) and low risk-trust $T^{\mathrm{risk}}{}_k$ (unreliable risk assessments), or vice versa. The two trust channels operate independently.

Initialization and cooling period. New agents enter with $\sigma_j^{\mathrm{risk},2}(0) = \sigma_0^{\mathrm{risk},2}$ (a prior reflecting baseline uncertainty about risk-claim quality), producing initial risk-trust $T^{\mathrm{risk}}{}_j(0) = 1/(1 + \beta_{\mathrm{risk}} \cdot \sigma_0^{\mathrm{risk},2})$. The effective cooling period is $\tau_{\mathrm{risk}} = \lceil \log\delta / \log\alpha_{\mathrm{risk}} \rceil$ (same formula as the behavioral cooling period, Equation 71 of Lasser [2026a]).

Proposition 2 (Risk-Trust L^2 Convergence)**.** *Let $X_j(t) = \|r_j^{\mathrm{risk}}(t)\|^2$ and assume a_j's risk-claim strategy is stationary with $\mu_j^{\mathrm{risk}} = \mathbb{E}[X_j] < \infty$ and $\nu_j^{\mathrm{risk}} = \mathrm{Var}[X_j] < \infty$. Then the cumulative*

inconsistency estimator satisfies:

$$\lim_{t\to\infty} \mathbb{E}\big[\sigma_j^{\mathrm{risk},2}(t)\big] = \mu_j^{\mathrm{risk}} \qquad \textit{(asymptotic unbiasedness)} \tag{7}$$

$$\limsup_{t\to\infty} \mathrm{Var}\big[\sigma_j^{\mathrm{risk},2}(t)\big] \le \frac{1-\alpha_{\mathrm{risk}}}{1+\alpha_{\mathrm{risk}}} \cdot C_{\mathrm{dep}} \cdot \nu_j^{\mathrm{risk}} \qquad \textit{(bounded stationary variance)} \tag{8}$$

where $C_{\mathrm{dep}} = 1$ under i.i.d. squared residuals, and $C_{\mathrm{dep}}(\alpha_{\mathrm{risk}}) = 1 + 2\sum_{h=1}^{\infty} \alpha_{\mathrm{risk}}^h \cdot \mathrm{Corr}(X_j(t), X_j(t+h))$ under weak dependence with summable autocovariance. The α_{risk}^h weights arise from the EWMA's geometric weighting: the cross-terms in the variance expansion carry the product of the EWMA coefficients at lag h. Since $|\mathrm{Corr}| \le 1$ and $\sum_h \alpha_{\mathrm{risk}}^h = \alpha_{\mathrm{risk}}/(1-\alpha_{\mathrm{risk}})$, we have the universal bound $C_{\mathrm{dep}} \le (1+\alpha_{\mathrm{risk}})/(1-\alpha_{\mathrm{risk}})$. Consequently, the variance estimator $\sigma_j^{\mathrm{risk},2}(t)$ concentrates in L^2 around μ_j^{risk}. The risk-trust factor ${T^{\mathrm{risk}}}_j(t)$ concentrates around the nominal fixed point ${T^{\mathrm{risk}}}_j^* = 1/(1+\beta_{\mathrm{risk}}\mu_j^{\mathrm{risk}})$ in distribution (via the continuous mapping theorem), with the caveat that $\mathbb{E}[{T^{\mathrm{risk}}}_j(t)]$ may exceed ${T^{\mathrm{risk}}}_j^*$ by a Jensen bias term of order $O(\beta_{\mathrm{risk}}^2 \cdot \mathrm{Var}[\sigma_j^{\mathrm{risk},2}])$ due to the concavity of $x \mapsto 1/(1+\beta_{\mathrm{risk}}x)$. The deviation is controlled by α_{risk}: as $\alpha_{\mathrm{risk}} \to 1$, the stationary variance vanishes, the Jensen bias vanishes, and ${T^{\mathrm{risk}}}_j(t) \to {T^{\mathrm{risk}}}_j^*$ in L^2. Structurally accurate reporters (small μ_j^{risk}) concentrate around high ${T^{\mathrm{risk}}}_j^*$; alarmists and chronic underreporters concentrate around low ${T^{\mathrm{risk}}}_j^*$.

Proof. Unrolling the EWMA recursion gives $\sigma_j^{\mathrm{risk},2}(t) = (1-\alpha_{\mathrm{risk}})\sum_{s=0}^{t-1} \alpha_{\mathrm{risk}}^s X_j(t-s) + \alpha_{\mathrm{risk}}^t \sigma_j^{\mathrm{risk},2}(0)$. Taking expectations under stationarity, $\mathbb{E}[\sigma_j^{\mathrm{risk},2}(t)] = (1-\alpha_{\mathrm{risk}})\sum_{s=0}^{t-1} \alpha_{\mathrm{risk}}^s \mu_j^{\mathrm{risk}} + \alpha_{\mathrm{risk}}^t \sigma_j^{\mathrm{risk},2}(0) \to \mu_j^{\mathrm{risk}}$ as $t \to \infty$, establishing Equation (7). Under the i.i.d. assumption,

$$\begin{aligned}\mathrm{Var}[\sigma_j^{\mathrm{risk},2}(t)] &= (1-\alpha_{\mathrm{risk}})^2 \sum_{s=0}^{t-1} \alpha_{\mathrm{risk}}^{2s} \cdot \nu_j^{\mathrm{risk}} + o(1) \\ &\longrightarrow \frac{(1-\alpha_{\mathrm{risk}})^2}{1-\alpha_{\mathrm{risk}}^2} \cdot \nu_j^{\mathrm{risk}} = \frac{1-\alpha_{\mathrm{risk}}}{1+\alpha_{\mathrm{risk}}} \cdot \nu_j^{\mathrm{risk}},\end{aligned}$$

giving $C_{\mathrm{dep}} = 1$ in Equation (8). Under weak dependence with summable autocovariance, the cross-terms in the EWMA variance expansion carry α_{risk}^h weights from the geometric kernel, giving $C_{\mathrm{dep}}(\alpha_{\mathrm{risk}}) = 1 + 2\sum_{h=1}^{\infty} \alpha_{\mathrm{risk}}^h \cdot \mathrm{Corr}(X_j(t), X_j(t+h))$, which is finite because α_{risk}^h decays geometrically and the autocovariance is summable. The concentration statement for $\sigma_j^{\mathrm{risk},2}(t)$ around μ_j^{risk} in L^2 follows directly. For ${T^{\mathrm{risk}}}_j(t)$, the continuous mapping theorem applied to $x \mapsto 1/(1+\beta_{\mathrm{risk}}x)$, which is Lipschitz on $[0,\infty)$, gives convergence in distribution to ${T^{\mathrm{risk}}}_j^*$. The Jensen bias arises because $\mathbb{E}[f(X)] \ge f(\mathbb{E}[X])$ for concave f; a second-order Taylor expansion gives the bias as $O(\beta_{\mathrm{risk}}^2 \cdot \mathrm{Var}[\sigma_j^{\mathrm{risk},2}])$, which vanishes as $\alpha_{\mathrm{risk}} \to 1$. ■

Tracking versus convergence: why L^2 and not almost sure. Fixed-α EWMA does not converge almost surely to a deterministic limit, and for an adversarial-tracking setting this is the correct behavior. Each new residual contributes weight $(1-\alpha_{\mathrm{risk}})$ regardless of how much history has accumulated, so the estimator never collapses to a point mass and retains nonzero stationary variance. A Robbins-Monro stochastic approximation with step size $\alpha_t \to 0$ satisfying $\sum_t \alpha_t = \infty$ and $\sum_t \alpha_t^2 < \infty$ [Robbins and Monro, 1951] would converge almost surely to μ_j^{risk}, but only at the cost of losing the ability to track a non-stationary adversary whose strategy changes mid-trajectory. Risk-claim quality and scorpion behavior are both moving targets: a cooperative reporter can become systematically biased, an alarmist can calibrate, and a scorpion can change strategy in response to detection pressure. Past accuracy does not guarantee future accuracy under a changing world model $\mathcal{W}$, so the tracking property is load-bearing. The stationary-variance bound in Equation (8) is what the scorpion detection threshold (Section 4) is calibrated against, so the L^2 form is not just weaker than almost sure — it is the form the downstream results actually use.

Interaction with the Wamura problem. The EWMA dynamics converge on *structural verification quality*, not on *outcome correctness*. An agent whose risk claims are structurally plausible

(the pathway exists in $\mathcal{M}$, the probability and magnitude estimates are calibrated against the actor's own model) accumulates low $\|r^{\text{risk}}\|^2$ and converges to high ${T^{\text{risk}}}_j^*$ regardless of whether the risk materializes. This partially resolves the Wamura problem: risk-trust converges even when the counterfactual is never observed, because convergence depends on structural agreement, not outcome observation. The resolution is partial because an agent whose claims are structurally plausible but whose probability estimates are systematically biased (always 10% higher than the actor's model) accumulates moderate $\|r^{\text{risk}}\|^2$ and converges to moderate ${T^{\text{risk}}}_j^*$, reflecting the bias without resolving it.

3.3 Independence-Preserving Risk Aggregation

The max operator in Equation 7 of Lasser [2026b] destroys independence information: 50 low-trust agents agreeing on a risk assessment are treated identically to one low-trust agent. We propose a trust-weighted log-odds aggregation that preserves the evidence-accumulation structure of independent reports.

Definition update from the companion paper. The companion paper [Lasser, 2026b] defines $\hat{\mathcal{R}}_j$ as an expected-contraction product $T_j^{\text{risk}} \cdot p \cdot |\Delta\text{vol}_{\text{P}}|$, which is not a probability and cannot be passed through log-odds. We update the definition: $\hat{\mathcal{R}}_j$ and $\mathcal{R}_{\text{self}}$ hereafter denote posterior probabilities that the exercise pathway will be executed within the planning horizon, calibrated against the shared prior π_0 below. The contraction magnitude $|\Delta\text{vol}_{\text{P}}|$ becomes a separate quantity used by the verification gate (Section 3.4) and the structural verification residual (Definition 3). The canonical form of a communicated risk claim is the pair $(\hat{\mathcal{R}}_j, |\Delta\text{vol}_{\text{P}}{}_j^{\text{claimed}}|)$; the single-number product in Lasser [2026b] is superseded.

Definition 5 (Log-Odds Risk Aggregation)**.** *Let $\pi_0 \in (0,1)$ be a shared prior on the probability that the exercise pathway will be executed within the planning horizon, and let $\ell_0 = \log(\pi_0/(1-\pi_0))$ be its log-odds. The actor's self-assessment $\mathcal{R}_{\text{self}}$ and each communicated risk estimate $\hat{\mathcal{R}}_j$ from agent a_j are posterior probabilities derived from the shared prior π_0 plus their respective observations (definition update, see above). The aggregated posterior is:*

$$\log \frac{\mathcal{R}_{\text{agg}}}{1-\mathcal{R}_{\text{agg}}} = \ell_0 + \underbrace{\left(\log \frac{\mathcal{R}_{\text{self}}}{1-\mathcal{R}_{\text{self}}} - \ell_0\right)}_{\textit{actor's log-likelihood-ratio}} + \sum_j {T^{\text{risk}}}_j \cdot \underbrace{\left(\log \frac{\hat{\mathcal{R}}_j}{1-\hat{\mathcal{R}}_j} - \ell_0\right)}_{\textit{reporter } a_j\textit{'s log-likelihood-ratio}} \tag{9}$$

where each parenthesized term is the log-likelihood-ratio contributed by that source beyond the shared prior, obtained by subtracting ℓ_0 from their reported posterior log-odds. The $\mathcal{R}$ values are normalized to $[0,1]$ as probabilities of the exercise pathway being executed within the planning horizon. The shared prior π_0 is an operational coordination requirement: every reporter's posterior must be calibrated against the same π_0 for the subtraction $\log(\hat{\mathcal{R}}_j/(1-\hat{\mathcal{R}}_j)) - \ell_0$ to correctly extract the log-likelihood ratio. Establishing π_0 requires a protocol (e.g., broadcasting it as a system parameter at population initialization), not just a mathematical assumption. Under the neutral bootstrap prior $\pi_0 = 0.5$ ($\ell_0 = 0$), the prior-correction terms vanish and Equation (9) *reduces to a naive sum of posterior log-odds; under any other shared prior, the correction is required to avoid double-counting the prior.*

The log-odds aggregation has three advantages over the max operator:

1. **Agreement strengthens evidence.** Multiple independent low-trust agents reporting the same risk produce a collective signal proportional to the number of agreeing reporters, weighted by their individual ${T^{\text{risk}}}_j$ values.
2. **Trust-weighted log-linear opinion pool.** Equation (9) is a trust-tempered log-linear opinion pool [Genest and Zidek, 1986]: the aggregate log-odds is a weighted sum of the reporters' log-likelihood ratios plus the shared prior. Under the conditional-independence assumption (reporters' observations are independent given the true risk level and the shared prior π_0), setting all ${T^{\text{risk}}}_j = 1$ recovers the standard Bayesian posterior update for independent evidence sources. The trust weights ${T^{\text{risk}}}_j \in [0,1]$ temper each reporter's contribution in proportion to their structural-verification track record; this is not itself Bayes-optimal in general — exact Bayes-optimality would require a generative model of how

each reporter's noise depends on their history, which the framework does not assume — but it recovers Bayes-optimal behavior in the no-tempering limit and degrades gracefully for untrusted reporters, which is the behavior the aggregation rule is trying to achieve. The prior-correction terms are required regardless of tempering: without them, a naive sum of posterior log-odds would double-count π_0 once per reporter.

3. **Graceful degradation.** When a single agent has ${T^{\text{risk}}}_j \gg \sum_{k \neq j} {T^{\text{risk}}}_k$, the aggregation approximates max behavior: the dominant reporter's assessment controls the aggregate. The max operator emerges as a limiting case, not an imposed design choice.

Bootstrap interaction. During bootstrap, all ${T^{\text{risk}}}_j$ values start at the neutral prior. Under log-odds aggregation, n agents reporting the same risk at neutral trust produce a collective signal $n \times {T^{\text{risk}}}_{\text{neutral}}$ times the individual log-odds, proportional to n. Under the max operator, the same n agents produce only ${T^{\text{risk}}}_{\text{neutral}}$ times the maximum individual log-odds, independent of n. The log-odds aggregation therefore learns faster during bootstrap, precisely when the actor's own model is least reliable and external evidence is most valuable.

3.4 Aggregation as Attention, Not Action

The log-odds aggregation amplifies agreement among reporters, which is correct when reporters are independent but exploitable by coordinated conspirators: n agents who agree in advance on a fabricated risk claim produce a collective signal proportional to n, potentially overwhelming the actor's self-assessment and triggering restriction of a capability that was never actually dangerous. This is the risk-communication analogue of report-bombing: coordinated low-quality reports exploiting an aggregation mechanism designed for independent evidence.

Any fixed correlation-detection heuristic is game-able once the adversary knows the parameters — a standard observation in adversarial detection settings — and the ROC trade-off is unfavorable in this domain because both error types are costly: false positives (restricting a safe capability) and false negatives (ignoring a real risk) both cause vol_{P}-contraction. We therefore adopt the activation energy principle from the foundational paper [Lasser, 2026a], applied to risk-claim processing.

The two-gate architecture. The risk aggregation $\mathcal{R}_{\text{agg}}$ (Definition 5) does not directly trigger restriction. It produces a *salience signal*: the aggregated risk level determines the priority at which a capability tuple enters the actor's investigation queue. Action requires passing two independent gates:

1. **Salience gate.** The aggregated risk $\mathcal{R}_{\text{agg}}(S; a_k, t)$ exceeds an attention threshold θ_{risk}, placing the capability tuple in the investigation queue.

2. **Verification gate.** The actor invokes the forward causal risk-evaluation procedure (Section 4.3 of Lasser [2026b]) on its own SCM $\mathcal{M}$ to independently confirm that the claimed exercise pathway $\mathcal{P}$ exists in the actor's causal model, calibrate the probability, and verify the contraction magnitude. This is a forward causal query — "if $\mathcal{P}$ were executed, what would happen?" — using the same pathway-conditional assessment $\big(p_{\mathcal{W}}(\mathcal{P}), |\Delta\text{vol}_{\text{P}\,\mathcal{W}}(\mathcal{P})|\big)$ as the structural verification residual (Definition 3). It is distinct from the retrospective causal contraction attribution (Definition 2), which decomposes an *observed* contraction across agents after the fact. Retrospective attribution has no counterfactual access to hypothetical pathways that were never executed; forward risk evaluation does, and this is the mechanism the verification gate requires. The gate is the activation energy: the actor must reconstruct the risk through its own causal model before acting.

Neither gate alone is sufficient. A coordinated report-bomb raises salience (gate 1) but fails independent verification (gate 2) because the fabricated pathway does not exist in the actor's causal model. A genuine risk reported by a single low-trust agent may fail gate 1 (low salience) but would pass gate 2 if the actor investigated. The two-gate architecture trades a bounded false-negative rate (genuine risks from low-trust reporters may be investigated slowly) for strong robustness against coordinated manipulation (fabricated risks cannot bypass independent verification regardless of how many conspirators report them).

Why heuristic correlation detection is insufficient. A fixed correlation metric (temporal clustering, behavioral similarity among reporters, trust-history patterns) is game-able by a sufficiently informed adversary: adversaries who know the detection threshold can space their reports, diversify their trust-building histories, and avoid clustering. The two-gate architecture avoids this arms race by not relying on detecting coordination at all. The defense is the actor's own intelligence applied through the causal model, not a statistical filter on the report stream.

4 Causal Scorpion Detection

The foundational paper's Proposition 2 [Lasser, 2026a] detects scorpions through two channels: contraction attribution (correlational) and deception detection (behavioral trust decay). This section upgrades channel (a) from correlational to causal, using the SCM of Section 2, and provides bounds on non-stationary scorpion evasion.

4.1 Causal Detection

Proposition 3 (Causal Scorpion Detection)**.** *Under the SCM $\mathcal{M}$ (Definition 1), a GFM actor can detect scorpion agents through two upgraded channels:*

(a) Causal contraction attribution. $\mathrm{CA}(a_j, t)$ *(Definition 2) replaces the correlational $\hat{\Delta}$ signal from Proposition 2(a) of Lasser [2026a]. Within each stationary epoch of a_j's strategy, the causal-attribution sequence $\mathrm{CA}(a_j, 1), \mathrm{CA}(a_j, 2), \ldots$ is i.i.d. $\mathcal{N}(\mu_j, \sigma^2_{\mathrm{do}})$ with unknown mean μ_j and known variance σ^2_{do}. (The Gaussian assumption is a modeling choice; for non-Gaussian attributions the SPRT likelihoods below are replaced by their distributional counterparts and the $O(\cdot)$ rate constants change, but the structural comparison with the correlational detector is preserved.) The actor maintains two auxiliary EWMA statistics — a* signed-mean *estimator and a* variance *estimator — for monitoring:*

$$\mu_j^{\mathrm{causal}}(t) = \alpha \cdot \mu_j^{\mathrm{causal}}(t-1) + (1-\alpha) \cdot \mathrm{CA}(a_j, t) \tag{10}$$

$$\sigma_j^{\mathrm{causal},2}(t) = \alpha \cdot \sigma_j^{\mathrm{causal},2}(t-1) + (1-\alpha) \cdot \big(\mathrm{CA}(a_j, t) - \mu_j^{\mathrm{causal}}(t)\big)^2 \tag{11}$$

These statistics satisfy the same L^2 convergence guarantees as Proposition 2: asymptotically unbiased with stationary variance bounded by $\frac{1-\alpha}{1+\alpha}\sigma^2_{\mathrm{do}}$ for the signed mean. They do not converge almost surely to a deterministic limit, and we deliberately do not use them as the flag condition.

The flag condition is instead a sequential probability ratio test (SPRT) [Wald, 1945] on the causal-attribution sequence, testing the least-favorable simple hypotheses $H_0 : \mu_j = 0$ (non-scorpion) versus $H_1 : \mu_j = -\epsilon$ (scorpion with minimum detectable contraction magnitude $\epsilon > 0$). The composite problem ($\mu_j \geq 0$ vs. $\mu_j \leq -\epsilon$) is reduced to this simple pair by the standard least-favorable-distribution argument: Type I error is maximized under the null distribution closest to the alternative ($\mu_j = 0$), and Type II error is maximized under the alternative closest to the null ($\mu_j = -\epsilon$), so error guarantees for the simple pair bound the composite problem from above. The actor accumulates the log-likelihood ratio $\Lambda_j(t) = \sum_{s=1}^{t} \log\big(p_{H_1}(\mathrm{CA}(a_j, s))/p_{H_0}(\mathrm{CA}(a_j, s))\big)$ and applies a two-boundary decision rule: flag a_j as a scorpion when $\Lambda_j(t) \geq \log((1-\beta_{\mathrm{err}})/\alpha_{\mathrm{err}})$ (upper boundary), or provisionally clear a_j for the current epoch when $\Lambda_j(t) \leq \log(\beta_{\mathrm{err}}/(1-\alpha_{\mathrm{err}}))$ (lower boundary). Here α_{err} and β_{err} are the target Type I and Type II error rates for the simple pair. A provisional clearance is epoch-local: the SPRT resets at the next epoch boundary, reflecting the possibility that a non-scorpion agent may become one. Under this decision rule with Gaussian likelihoods, the per-observation expected log-likelihood ratio under the true mean μ_j is $\mathbb{E}_{\mu_j}[\log(p_{H_1}/p_{H_0})] = \epsilon(|\mu_j| - \epsilon/2)/\sigma^2_{\mathrm{do}}$, giving expected sample size $\mathbb{E}[\tau_{\mathrm{detect}}] = O\big(\sigma^2_{\mathrm{do}}/[\epsilon(|\mu_j| - \epsilon/2)]\big)$ by Wald's identity. At the design point $|\mu_j| = \epsilon$ this reduces to $O(\sigma^2_{\mathrm{do}}/\epsilon^2)$; for $|\mu_j| \gg \epsilon$ it improves to $O(\sigma^2_{\mathrm{do}}/(\epsilon|\mu_j|))$. By Markov's inequality the detection probability within $k \cdot \mathbb{E}[\tau_{\mathrm{detect}}]$ observations is at least $1 - 1/k$ for any $k > 1$. (Under the Gaussian model, τ_{detect} follows an inverse Gaussian distribution whose tail decays exponentially [Wald, 1945], so the Markov bound is conservative; exponentially tight high-probability detection guarantees are available at the cost of distributional

specificity.) The structural advantage over the correlational detector (Proposition 1) is preserved: the causal detector operates on variance σ^2_{do} while the correlational detector operates on $\sigma^2 \geq \sigma^2_{\text{do}}$. The signed-mean EWMA is retained as a human-interpretable monitor of the estimator state; the SPRT statistic $\Lambda_j(t)$ is the load-bearing decision variable.

(b) ***Deception detection (unchanged).*** *The behavioral trust factor T_j detects agents whose reported capability changes diverge from observations, through the same prediction-residual dynamics as the foundational paper (Equations 25, 33, 38 of Lasser [2026a]). Causal attribution does not affect this channel: a deceptive agent accumulates prediction residuals regardless of whether the contraction is causally attributed.*

Proof. Channel (a): By the SCM, $\text{CA}(a_j, t)$ isolates a_j's causal contribution from confounders (Definition 2). Within a stationary epoch of a_j's strategy, $\{\text{CA}(a_j, t)\}$ is i.i.d. with mean μ_j and variance σ^2_{do}; for a scorpion $\mu_j \leq -\epsilon < 0$. The auxiliary EWMAs $\mu_j^{\text{causal}}(t)$ and $\sigma_j^{\text{causal},2}(t)$ inherit the L^2 guarantees of Proposition 2: asymptotically unbiased estimators of μ_j and σ^2_{do} with bounded stationary variance. They are not used for the detection decision, so we do not require almost-sure convergence.

The SPRT is applied to the least-favorable simple pair $H_0 : \mu_j = 0$ versus $H_1 : \mu_j = -\epsilon$ (see channel (a) above for the reduction from composite to simple hypotheses). Under the Gaussian model, the per-observation expected log-likelihood ratio under the true mean μ_j is $\mathbb{E}_{\mu_j}[\log(p_{H_1}/p_{H_0})] = \epsilon(|\mu_j| - \epsilon/2)/\sigma^2_{\text{do}}$. By Wald's identity, the expected detection time under true μ_j is $\mathbb{E}[\tau_{\text{detect}}] \approx \log((1-\beta_{\text{err}})/\alpha_{\text{err}}) \cdot \sigma^2_{\text{do}}/[\epsilon(|\mu_j| - \epsilon/2)]$, which at the design point $|\mu_j| = \epsilon$ reduces to $O(\sigma^2_{\text{do}}/\epsilon^2)$ and for $|\mu_j| \gg \epsilon$ improves to $O(\sigma^2_{\text{do}}/(\epsilon|\mu_j|))$. The SPRT Type I error for the simple pair is bounded by α_{err} and Type II by β_{err}; the least-favorable argument ensures these bounds hold for the composite problem.

Channel (b): The behavioral trust dynamics (Appendix B of Lasser [2026a]) operate on behavioral prediction residuals $r_j(t) = R_j(t) - \hat{R}_j(t)$, which are independent of causal attribution. A deceptive scorpion that misreports capability changes accumulates $r_j(t) \neq 0$ regardless of confounders, driving $T_j \to T_j^* < 1$. ■

4.2 Non-Stationary Scorpion Bounds

The foundational paper notes that "a non-stationary scorpion that adapts its strategy in response to detection pressure can evade both channels indefinitely" (Proposition 2 of Lasser [2026a]). Under the SCM, we can bound the adaptation rate required for evasion.

Proposition 4 (Evasion Bandwidth)**.** *A scorpion a_j with per-epoch mean causal contraction $|\mu_j|$ that changes strategy every τ_{adapt} steps produces at most τ_{adapt} observations per strategy epoch. Let $\bar{\tau}(\mu_j) = O\big(\sigma^2_{\text{do}}/[\epsilon(|\mu_j| - \epsilon/2)]\big)$ be the expected detection time per epoch from the SPRT under the Gaussian model with design parameter ϵ (Proposition 3). Then:*

(a) *If $\tau_{\text{adapt}} \geq k \cdot \bar{\tau}(\mu_j)$ for some $k > 1$, the scorpion is detected within each epoch with probability at least $1 - 1/k$ (by Markov's inequality on the SPRT stopping time).*

(b) *If $\tau_{\text{adapt}} < \bar{\tau}(\mu_j)$, the expected accumulated evidence at the epoch boundary is below the SPRT threshold. While this does not by itself guarantee evasion with any specific probability, it establishes that the expected number of observations is insufficient for detection, so the SPRT is unlikely to trigger within the epoch.*

Proof. Within each epoch of length τ_{adapt}, the scorpion's strategy is stationary and the causal-attribution sequence is i.i.d. $\mathcal{N}(\mu_j, \sigma^2_{\text{do}})$. The SPRT statistic accumulates log-likelihood-ratio evidence at expected rate $\epsilon(|\mu_j| - \epsilon/2)/\sigma^2_{\text{do}}$ per observation under the Gaussian model (Proposition 3), giving expected detection time $\bar{\tau}(\mu_j) = \sigma^2_{\text{do}} \cdot \log((1-\beta_{\text{err}})/\alpha_{\text{err}})/[\epsilon(|\mu_j| - \epsilon/2)]$.

(a) By Markov's inequality, $P(\tau_{\text{detect}} > k\bar{\tau}) \leq 1/k$ for any $k > 1$. When $\tau_{\text{adapt}} \geq k\bar{\tau}(\mu_j)$, the epoch provides enough observations that the SPRT crosses the detection threshold with probability at least $1 - 1/k$.

(b) An epoch boundary resets the SPRT statistic (the hypotheses are epoch-local). If $\tau_{\text{adapt}} < \bar{\tau}(\mu_j)$, the expected accumulated evidence at the epoch boundary is below the SPRT threshold. Since the number of observations is less than the expected detection time, the SPRT is unlikely to trigger within the epoch: the evidence accumulation rate is insufficient for reliable detection. ■

Interpretation. The evasion bandwidth $\tau^*_{\text{adapt}} = O\big(\sigma^2_{\text{do}}/[\epsilon(|\mu_j| - \epsilon/2)]\big)$ is the characteristic timescale: a scorpion that adapts faster than τ^*_{adapt} evades with non-negligible probability per epoch, while one that adapts slower than $k \cdot \tau^*_{\text{adapt}}$ is detected with probability at least $1 - 1/k$. A fast-adapting scorpion ($\tau_{\text{adapt}} \ll \tau^*_{\text{adapt}}$) evades but is constrained to small contractions per epoch (otherwise μ_j is large and τ^*_{adapt} shrinks). A slow-adapting scorpion ($\tau_{\text{adapt}} \gg \tau^*_{\text{adapt}}$) is detected within each epoch with high probability. The product $\mu_j \cdot \tau_{\text{adapt}}$ (total contraction per epoch) is bounded: a scorpion can either cause large damage slowly (and be detected) or small damage quickly (and evade), but not large damage quickly.

The time-averaged undetected contraction rate for an evading scorpion is bounded by $|\mu_j|$: within each epoch the per-step contraction is $|\mu_j|$, and the evasion constraint $\tau_{\text{adapt}} < \bar{\tau}(\mu_j)$ bounds the epoch length. The appendix proof (§A.1) establishes this via the SPRT accumulation rate. The key trade-off is controlled by the SPRT design parameter ϵ: a scorpion whose per-step contraction satisfies $|\mu_j| \leq \epsilon$ requires at least $\bar{\tau}(\epsilon) = O(\sigma^2_{\text{do}}/\epsilon^2)$ observations to detect (the worst case at the design point), while a scorpion with $|\mu_j| \gg \epsilon$ is detected faster in proportion to $|\mu_j|/\epsilon$. The SPRT design parameter ϵ therefore sets the resolution: reducing ϵ detects smaller contractions but requires more observations per epoch. The bound tightens as the causal model improves (σ_{do} decreases) and as the scorpion's adaptation budget τ_{adapt} grows: a scorpion that adapts frequently (small τ_{adapt}) can only afford very small per-step contractions, while a scorpion that adapts slowly can sustain larger contractions per step but for correspondingly fewer steps before detection.

5 Probabilistic Dependency Risk

The companion paper [Lasser, 2026b] bounds the worst-case contraction of single-substrate versus cross-substrate populations through a minimax analysis (Proposition 4). That bound is a one-time penalty on the value function: useful for showing diversification is structurally preferred, but not directly usable inside a discounted trajectory because it does not attach a per-step cost. This section converts the minimax bound into an expected per-step dependency-risk cost, then folds that cost into the discounted value function so diversification enters the horizon calculation as a first-class term rather than a static penalty.

The section has three parts. First, a probabilistic substrate threat model (Section 5.1) that decomposes dependency risk into event probability, target selection, and cascade effects. Second, an expected per-step risk comparison (Proposition 5) that sharpens the minimax bound into an *adversarial balance condition* $c > \max_i(f_i \cdot c_i)$ — a single inequality that gates when diversification strictly beats concentration and that explicitly admits contagion dynamics (pandemics, cascading failures, memetic attacks) as the motivating threat class. Third, a risk-adjusted trajectory analysis (Section 5.3) showing that the per-step cost compounds at the same discount rate as growth, so the diversification-versus-concentration decision becomes a direct comparison of two sums in the value function.

5.1 Substrate Threat Model

Definition 6 (Substrate Threat Model)**.** *A* substrate threat model *is a distribution over adversarial events, parametrized by:*

- p_{adv}*: per-step probability that a substrate-targeting event occurs.*
- $q_i = p(\mathcal{S}_i$ *targeted* $\mid$ *event occurs*$)$*: the conditional probability that substrate class* $\mathcal{S}_i$ *is targeted, given that an event occurs. For an uninformed adversary,* $q_i = 1/m$ *(uniform); for an optimal adversary,* $q_i = \mathbf{1}[i = \arg\max_j(|\Delta\text{vol}_{\text{P}}(\mathcal{S}_j)| \cdot c_j)]$*, i.e., the adversary targets the substrate with maximal* effective damage *(volume contribution times cascade fraction).*

- $c_i \in [0,1]$*: the* cascade fraction *for substrate $\mathcal{S}_i$, defined as the expected fraction of $\mathcal{S}_i$'s total volume lost when a substrate-targeting event hits $\mathcal{S}_i$. "Cascade" refers to the propagation dynamics — within-substrate impairment spreading from the initial strike through the SCM's structural equations — and "fraction" constrains c_i to $[0,1]$ as a proportion of $\mathcal{S}_i$'s volume. The cascade fraction aggregates the initial impairment and all subsequent within-substrate propagation: if a direct attack impairs 30% of the substrate and triggers a contagion that takes out another 40%, then $c_i = 0.7$. The homogeneous single-substrate baseline uses a single c in place of the per-substrate c_i.*

In this threat model $|\Delta\mathrm{vol}_\mathrm{P}(\mathcal{S}_i)|$ denotes substrate $\mathcal{S}_i$'s total contribution to $\mathrm{vol}_\mathrm{P}(G)$ *— i.e., the maximum possible contraction if $\mathcal{S}_i$ were fully eliminated — not the expected contraction from a targeting event. The expected contraction is $|\Delta\mathrm{vol}_\mathrm{P}(\mathcal{S}_i)| \cdot c_i$: substrate size times cascade fraction, with the two quantities varied independently. This convention matches Proposition 4 of Lasser [2026b], where $|\Delta\mathrm{vol}_\mathrm{P}(\mathcal{S}_i)|$ is the minimax-bound substrate size, and prevents double-counting when the cascade fraction is applied as a multiplier.*

The threat model decomposes dependency risk into three components: occurrence probability, target selection, and cascading effects. The minimax analysis of Lasser [2026b] assumes $p_\mathrm{adv} > 0$ (non-negligible threat), q_i chosen adversarially (worst case), and full cascade within the targeted substrate (all agents on that substrate are impaired). The probabilistic model relaxes each of these: p_adv is estimated from the actor's world model, q_i reflects the actor's beliefs about threat targeting, and cascading effects are modeled through the SCM's structural equations.

5.2 Expected Per-Step Dependency Risk

Notation update from the companion paper. The companion paper [Lasser, 2026b] defines $\mathcal{R}_\mathrm{dep}$ as a discounted pathway sum over the full planning horizon. In this section, $\mathcal{R}_\mathrm{dep}^\mathrm{single}$ and $\mathcal{R}_\mathrm{dep}^\mathrm{mixed}$ denote the *per-step expected dependency-risk cost* under the substrate threat model (Definition 6): a single-step expected contraction, not a discounted sum. The per-step form is the input to the risk-adjusted trajectory comparison (Section 5.3), which recovers a discounted sum by standard geometric-series aggregation.

Proposition 5 (Expected Risk Comparison)**.** *Under the substrate threat model (Definition 6), the expected per-step dependency risk for each strategy is:*

$$\mathbb{E}[\mathcal{R}_\mathrm{dep}^\mathrm{single}] = p_\mathrm{adv} \cdot \mathrm{vol}_\mathrm{P}(G) \cdot c \tag{12}$$

$$\mathbb{E}[\mathcal{R}_\mathrm{dep}^\mathrm{mixed}] = p_\mathrm{adv} \cdot \sum_i q_i \cdot |\Delta\mathrm{vol}_\mathrm{P}(\mathcal{S}_i)| \cdot c_i \tag{13}$$

where c and c_i are the cascade fractions from Definition 6. For a single-substrate population ($m = 1$), any substrate-targeting event affects a fraction c of the entire population. For a cross-substrate population ($m \geq 2$):

$$\mathbb{E}[\mathcal{R}_\mathrm{dep}^\mathrm{single}] - \mathbb{E}[\mathcal{R}_\mathrm{dep}^\mathrm{mixed}] = p_\mathrm{adv} \cdot \left(\mathrm{vol}_\mathrm{P}(G) \cdot c - \sum_i q_i \cdot |\Delta\mathrm{vol}_\mathrm{P}(\mathcal{S}_i)| \cdot c_i \right) \tag{14}$$

Under ***worst-case adversarial targeting*** *($q_i = \mathbf{1}[i = \arg\max_j(|\Delta\mathrm{vol}_\mathrm{P}(\mathcal{S}_j)| \cdot c_j)]$), this gap is positive whenever $p_\mathrm{adv} > 0$, the population satisfies substrate capability non-redundancy (Proposition 4 of Lasser [2026b]), and the* ***adversarial balance condition*** *$c > \max_i(f_i \cdot c_i)$ holds, where $f_i = |\Delta\mathrm{vol}_\mathrm{P}(\mathcal{S}_i)|/\mathrm{vol}_\mathrm{P}(G)$ is substrate $\mathcal{S}_i$'s share of total volume. The condition is both sufficient and necessary under worst-case targeting; for non-adversarial q_i (e.g., uniform), the condition is sufficient but not necessary.*

Proof. Under an adversarial target-selection model, $q_i = \mathbf{1}[i = i^*]$ where $i^* = \arg\max_j(|\Delta\mathrm{vol}_\mathrm{P}(\mathcal{S}_j)| \cdot c_j)$. Substituting into Equation (13) gives

$$\mathbb{E}[\mathcal{R}_\mathrm{dep}^\mathrm{mixed}] = p_\mathrm{adv} \cdot \max_i\big(|\Delta\mathrm{vol}_\mathrm{P}(\mathcal{S}_i)| \cdot c_i\big) = p_\mathrm{adv} \cdot \mathrm{vol}_\mathrm{P}(G) \cdot \max_i(f_i \cdot c_i),$$

while $\mathbb{E}[\mathcal{R}_\mathrm{dep}^\mathrm{single}] = p_\mathrm{adv} \cdot \mathrm{vol}_\mathrm{P}(G) \cdot c$. The gap is positive iff $c > \max_i(f_i \cdot c_i)$. Substrate capability non-redundancy (Proposition 4 of Lasser [2026b]) ensures $f_i < 1$ strictly for every i, so the condition is non-vacuous for any $c > 0$: some non-trivial region of (f_i, c_i) space satisfies it and some does not, and the proposition is a sharp characterization of which region. ■

On contagion dynamics. The adversarial balance condition admits contagions by construction, which is essential because contagion-driven threats — pandemics, cascading infrastructure failures, financial contagion, memetic attacks — are precisely the class where per-substrate cascade fractions can approach saturation ($c_i \to 1$) on the targeted substrate. Smaller substrates are expected to reach sigmoid saturation faster than larger ones because contagion propagation time scales sub-linearly with population size in standard epidemiological models [Anderson and May, 1991], so $c_i > c$ is a plausible assumption for the substrates in a diversified partition. The gap remains positive as long as each substrate's volume share f_i is small enough to compensate.

Two instructive limits bracket the regime of interest:

- **Full-saturation contagion** ($c_i = c = 1$): the condition reduces to $1 > f_{\max}$, which is exactly the substrate-capability-non-redundancy assumption from Lasser [2026b]. No additional structure is required.
- **Unbalanced partition under contagion amplification** ($c_i > c$, $f_{\max}$ large): the condition can fail. For example, $c = 0.5$, $c_{\max} = 1.0$, $f_{\max} = 0.6$ gives $0.5 \not> 0.6$ and the gap reverses. In this regime diversification provides no dependency-risk advantage over the single-substrate baseline: the adversary still targets the dominant substrate, and the contagion saturates it more thoroughly than in the homogeneous case. The trust model should detect this regime and avoid unbalanced partitions when contagion risks dominate; the bound flags the edge case rather than assuming it away.

The analysis assumes cascade confinement: c_i measures only the fraction of substrate $\mathcal{S}_i$ lost when $\mathcal{S}_i$ is targeted, not propagation into other substrates. Cross-substrate cascades — a memetic attack on $\mathcal{S}_1$ that triggers a cascade in $\mathcal{S}_2$ through inter-substrate communication channels — are outside the scope of the current bound and require a separate treatment beyond the current SCM.

Corollary 5.1 (Balanced-Partition Substrate Count)**.** *Under a balanced substrate partition with $f_i = 1/m$ for all i, the adversarial balance condition of Proposition 5 is equivalent to*

$$m > \frac{c_{\max}}{c}, \tag{15}$$

where $c_{\max} = \max_i c_i$ is the worst-case per-substrate cascade fraction.

Proof. With $f_i = 1/m$, the adversarial balance condition $c > \max_i(f_i \cdot c_i)$ reduces to $c > \max_i(c_i/m) = c_{\max}/m$, equivalent to Equation (15). ■

Corollary 5.1 sharpens the minimax bound of Lasser [2026b], which establishes only that $m \geq 2$ strictly dominates $m = 1$. The minimax statement is the special case $c_{\max} = c$ of Equation (15): cascade intensity unchanged by partition size gives $m > 1$, recovering $m \geq 2$. Weakening either parameter raises the required substrate count:

- **Moderate contagion** ($c = 0.5$, $c_{\max} = 1.0$): $m > 2$, so $m \geq 3$. Binary diversification is insufficient.
- **High contagion** ($c = 0.1$, $c_{\max} = 1.0$): $m > 10$, so $m \geq 11$. The substrate-count requirement grows inversely with the homogeneous cascade fraction.
- **Low contagion** ($c \approx c_{\max} \approx 0.3$, localized failure modes): $m > 1$, recovering the minimax bound.

Practically: the minimum number of substrates required for diversification to strictly help depends on the measured contagion intensity, not just on the structural claim that $m \geq 2$. A binary partition is correct for threat classes where the homogeneous cascade already saturates ($c \to 1$), but underprovisions for contagion-driven threats where smaller substrates saturate faster than the homogeneous baseline. For unbalanced partitions, the operational form $f_{\max} < c/c_{\max}$ is the direct test: it bounds the largest substrate's share rather than specifying a count, and is the condition the trust model should check when assessing a given partition's dependency-risk posture.

5.3 Risk-Adjusted Trajectory Comparison

The per-step expected risk gap compounds at the same rate as growth, unlike the one-time minimax penalty in Lasser [2026b]. Under the assumption that Δ_{risk} is stationary across the trajectory — that

is, p_{adv}, the per-substrate shares f_i, and the cascade fractions c_i are time-invariant in expectation — the geometric series sums to the standard discounted form:

$$V_\gamma^{\mathrm{div,risk}} - V_\gamma^{D,\mathrm{risk}} = \frac{r_{\mathrm{ext}} + \Delta_{\mathrm{risk}}}{1-\gamma} - \Delta_0 \tag{16}$$

where $\Delta_{\mathrm{risk}} = \mathbb{E}[\mathcal{R}_{\mathrm{dep}}^{\mathrm{single}}] - \mathbb{E}[\mathcal{R}_{\mathrm{dep}}^{\mathrm{mixed}}] > 0$ is the per-step risk gap from Proposition 5. The threshold for diversity dominance becomes:

$$\gamma_{\mathrm{risk}}^* = 1 - \frac{r_{\mathrm{ext}} + \Delta_{\mathrm{risk}}}{\Delta_0} \tag{17}$$

The risk gap Δ_{risk} enters additively with r_{ext}, effectively increasing the external contribution by the per-step risk mitigation value of cross-substrate diversity. When the risk gap is non-stationary — e.g., the threat landscape evolves, or the substrate partition itself changes during the trajectory — the closed-form $(1-\gamma)^{-1}$ denominator becomes an approximation, and the exact value is $\sum_{t=0}^{\infty} \gamma^t \cdot \Delta_{\mathrm{risk}}(t)$ computed from a time-varying risk profile. The trust model's standard non-stationarity assumption (agents and threats evolve on timescales slower than the discount horizon) makes the stationary approximation adequate for planning; rapidly-varying threat landscapes require a finer-grained value-iteration calculation instead of the closed form.

Quantitative impact. For a population with $p_{\mathrm{adv}} = 0.01$ (1% per-step probability of substrate-targeting event) and cascade fraction $c = 0.5$ (half the targeted substrate's capabilities are lost in a typical event), $\Delta_{\mathrm{risk}} = 0.01 \times \Delta_{\mathrm{div}} \times 0.5$ where Δ_{div} is the substrate diversification advantage from Proposition 4 of Lasser [2026b]. If Δ_{div} is 10% of $\mathrm{vol}_{\mathrm{P}}(G)$, then $\Delta_{\mathrm{risk}} \approx 0.0005 \times \mathrm{vol}_{\mathrm{P}}(G)$: the per-step risk cost of single-substrate concentration is 0.05% of total volume per step. Over the full planning horizon ($1/(1-\gamma)$ steps), this compounds to $0.05\% \times \mathrm{vol}_{\mathrm{P}}(G)/(1-\gamma)$, which for $\gamma = 0.99$ is $5\% \times \mathrm{vol}_{\mathrm{P}}(G)$—substantial.

Interaction with the horizon paper's cooperative-novelty result. The per-step risk gap Δ_{risk} is amplified by the cross-substrate cooperative capabilities analyzed in Lasser [2026b]: any cooperative capability requiring agents on both substrates is destroyed when either substrate is targeted, so $|\Delta\mathrm{vol}_{\mathrm{P}}(\mathcal{S}_i)|$ includes those cooperative losses on top of the individual-capability loss on $\mathcal{S}_i$. Under the horizon paper's cooperative-novelty result, these cross-substrate cooperative capabilities are the primary growth channel, so their destruction represents a disproportionate fraction of the total contraction during a substrate-targeting event. The cooperative novelty and dependency-risk analyses reinforce each other: domination is costly both because it eliminates the cross-substrate cooperative growth channel *and* because it concentrates adversarial exposure to the single surviving substrate [Lasser, 2026b].

Summary. The adversarial balance condition $c > \max_i(f_i \cdot c_i)$ converts the companion paper's minimax bound into an expected per-step cost Δ_{risk} that compounds as $\Delta_{\mathrm{risk}}/(1-\gamma)$ in the discounted value function, making dependency risk a first-class term in the horizon calculation rather than a one-time structural penalty. The condition is explicit about contagion dynamics and inverts (Corollary 5.1) into a quantitative bound on the required substrate count: $m > c_{\max}/c$ for balanced partitions, or $f_{\max} < c/c_{\max}$ for arbitrary partitions. This sharpens the minimax $m \geq 2$ into a measurement-driven requirement that scales with contagion intensity. The regime where the condition fails — unbalanced partitions under contagion amplification — is a regime where diversification genuinely does not help, which the trust model can detect and route around.

6 Discussion

6.1 Limitations

SCM structure learning as channel optimization, not prerequisite. The causal attribution channel of Section 2 requires the actor to know or learn the causal DAG over agent actions and capability changes. Standard observational structure learning [Spirtes et al., 2000] is inadequate for the GFM setting for three distinct reasons. First, the data-generating process is adversarial: scorpions can strategically align their actions with observed noise to mask their causal role, so classical conditional-independence tests can be gamed by an adversary who knows which tests will be run.

Standard structure learning does not have a threat model. Second, the agent cardinality is non-fixed: new agents can appear, existing agents can delegate, and a group can coordinate and behave as a single effective actor, so the variable set itself evolves. Latent-variable discovery methods (FCI, RFCI) assume a fixed variable cardinality and do not naturally handle this. Third, the actor cannot typically intervene on other agents: interventional causal discovery relies on actual interventions, which a GFM actor can simulate inside its SCM but cannot enforce in the world. This rules out the entire class of interventional discovery methods and leaves only observational methods, which are strictly weaker under the adversarial conditions above.

Because causal attribution is one channel among several (Section 2.4), and capability gating via the poset of Lasser [2026c] and Lasser [2026b] is the strongest and most difficult to fake channel (relative to a trusted, sufficiently complete capability graph), structure learning becomes an *optimization target* for one channel's contribution rather than a *prerequisite* for attribution at all. When the SCM is poorly learned, the causal channel degrades, the other channels are not directly degraded by the SCM misspecification, and the multi-channel aggregate reweights toward the structurally-grounded channels. The system does not collapse; it falls back on the trust and capability machinery of Lasser [2026a], Lasser [2026c], and Lasser [2026b].

The structural assumptions underlying Proposition 1(c) are partially testable through a residual-covariance diagnostic: if $\mathrm{Cov}(\mathrm{CA}(a_j), \mathrm{CA}(a_k) \mid G_t)$ exceeds the level the current SCM predicts, the SCM is misspecified — possible causes include a missing edge, latent common causes, exogenous-orthogonality failure, non-additivity, or nonstationarity — and the causal channel's reliability should degrade accordingly. Integrating such a diagnostic into a calibrated SCM-confidence scalar $c_{\mathcal{M}}(t) \in [0, 1]$ that weights the causal channel's contribution to the multi-channel aggregate — so declining sufficiency down-weights causal attribution without burning specific agents' trust — is an architectural direction we identify but do not formalize here. The full adversarial-robust structure-learning problem, including the calibration procedure and its interaction with the behavioral and capability channels, is deferred to a companion paper.

Risk-trust residual dependence on the actor's model. The pathway-conditional structural verification residual (Definition 3) measures divergence between the claim and the actor's own assessment of the claimed pathway. If the actor's world model is itself poorly calibrated, the residual reflects model error rather than claim quality. An agent whose risk assessments are accurate but disagree with a miscalibrated actor will accumulate high $\|r^{\mathrm{risk}}\|^2$ and concentrate around low $T^{\mathrm{risk}*}_j$ — a false negative. This is mitigated by the self-trust mechanism of Lasser [2026a]: an actor with a poorly calibrated model has low T_s and high learning rate, which accelerates model correction. But during the correction period, risk-trust assessments may be unreliable.

Log-odds aggregation and reporter correlation. The trust-weighted log-linear opinion pool (Definition 5) recovers Bayesian updating in the no-tempering limit under conditional independence of agents' observations given the true risk level. If reporters' assessments are correlated — they share information sources, or one reporter's assessment influences others — the aggregation overweights the shared signal. In practice, partial independence is the norm: agents with different observation channels and different world models produce partially independent assessments, and the log-linear pool is an approximation whose error grows with the correlation. Characterizing the approximation error under partial independence, and deriving a correction term from a reporter-network model, is deferred to future work.

7 Conclusion

The preceding GFM papers define what to maximize ($\mathrm{vol}_{\mathrm{P}}(G)$), prove structural safety properties, and establish anti-monopolar pressure. This paper addresses how the actor *reasons about other agents*: upgrading every evaluation mechanism from statistical to causal.

Causal attribution (Section 2) upgrades the foundational paper's correlational scorpion detection to causal identification through do-calculus counterfactuals. The convergence advantage over correlational detection is proportional to confounding variance, which is substantial in populations with many simultaneously acting agents. Non-stationary scorpion bounds (Section 4) characterize the evasion bandwidth: how fast a scorpion must adapt to remain undetected, and the maximum undetected contraction rate.

Risk-trust dynamics (Section 3) provide formal EWMA update equations for $T^{\mathrm{risk}}{}_j$, with a two-gate architecture separating attention allocation (log-odds aggregation) from action decisions (the actor's own causal verification). This prevents coordinated report-bombing while preserving sensitivity to genuine independent agreement. Probabilistic dependency risk (Section 5) converts minimax substrate bounds into per-step expected costs that compound at the same rate as growth, providing the quantitative risk model needed for risk-adjusted planning.

Together with the foundational framework [Lasser, 2026a], the computable poset [Lasser, 2026c], and the horizon-aware objective [Lasser, 2026b], these results narrow the gap between the GFM framework's structural guarantees and the mechanisms a concrete agent would need to realize them. The results here are conditional: the causal convergence advantage (Proposition 1(c)) relies on causal sufficiency, additive structure, and exogenous orthogonality of the SCM; the Bayesian interpretation of log-odds aggregation (Definition 5) relies on conditional independence given a shared prior; the dependency-risk gap (Proposition 5) relies on cascade confinement within substrates and the adversarial balance condition; and the evasion bound (Proposition 4) relies on a stationary-epoch adversary model. Each assumption is explicit, falsifiable, and load-bearing. The remaining work is both theoretical (relaxing each assumption and characterizing what happens when it fails) and empirical: instantiating these computations in a concrete environment and checking whether the predicted alignment properties hold under real conditions.

Author Contributions

Teague Lasser identified the statistical-to-causal upgrade as the unifying contribution, proposed the log-odds risk aggregation and the two-gate attention/verification architecture, directed all revisions, and made final editorial decisions. Responsible for the paper's intellectual direction and all claims made.

Claude Opus 4.6 (Anthropic) drafted the formal exposition: the structural causal model for capability dynamics, the do-calculus contraction attribution, the L^2 risk-trust EWMA dynamics, the pathway-conditional structural verification residual, the SPRT-based causal scorpion detection and evasion bandwidth bounds, and the probabilistic dependency-risk model with the adversarial balance condition.

GPT 5.4 (OpenAI) served as technical reviewer, identifying formal gaps in the risk-trust convergence claims, the causal attribution assumptions, the log-odds aggregation's cross-paper type consistency, and the dependency-risk quantitative analysis — in particular the need to replace the pointwise cascade-fraction assumption with the adversarial balance condition that admits contagion dynamics.

Transparency note. Both AI systems operated as tools under human direction. Neither system has continuity across sessions, cannot take responsibility for the work in the sense required by most venue authorship policies, and cannot respond to reviewer queries independently. They are listed as authors to accurately represent their contributions to the intellectual content of the paper, not to claim that they meet all criteria of traditional academic authorship. The corresponding author for all inquiries is Teague Lasser.

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A Proofs

Propositions 1 (Convergence Advantage), 2 (Risk-Trust Convergence), 3 (Causal Scorpion Detection), and 5 (Expected Risk Comparison) are proved in the main text. This appendix provides the remaining proofs.

A.1 Proof of Proposition 4 (Evasion Bandwidth)

Proof. Within each strategy epoch of length τ_{adapt}, the scorpion's behavior is stationary and the causal-attribution sequence $\{\text{CA}(a_j, t)\}$ is i.i.d. with mean $\mu_j < 0$ and variance σ^2_{do} (the same assumption as Proposition 3). The SPRT decision rule tests the least-favorable simple pair $H_0 : \mu_j = 0$ versus $H_1 : \mu_j = -\epsilon$ under the Gaussian model. The per-observation expected log-likelihood ratio under the true mean μ_j is $\mathbb{E}_{\mu_j}[\log(p_{H_1}/p_{H_0})] = \epsilon(|\mu_j| - \epsilon/2)/\sigma^2_{\text{do}}$. By Wald's identity [Wald, 1945], the expected detection time within a stationary epoch with actual mean μ_j is:

$$\bar{\tau}(\mu_j) = \mathbb{E}[\tau_{\text{detect}}] = \frac{\sigma^2_{\text{do}}}{\epsilon(|\mu_j| - \epsilon/2)} \cdot \log \frac{1 - \beta_{\text{err}}}{\alpha_{\text{err}}}.$$

Part (a): detection. By Markov's inequality, $P(\tau_{\text{detect}} > k\bar{\tau}(\mu_j)) \leq 1/k$ for any $k > 1$. When $\tau_{\text{adapt}} \geq k\bar{\tau}(\mu_j)$, the epoch provides enough observations that the SPRT crosses the detection threshold with probability at least $1 - 1/k$.

Part (b): evasion. On an epoch boundary the null and alternative hypotheses change (the new epoch has a new $(\mu_j, \sigma^2_{\text{do}})$), so the SPRT statistic is reset to zero: evidence accumulated under one pair of hypotheses is not informative about a different pair. The auxiliary EWMA monitors (Equations (10)–(11)) retain a fraction $\alpha^{\tau_{\text{adapt}}}$ of prior evidence as their output, but the SPRT decision rule itself is memoryless across regime changes. If $\tau_{\text{adapt}} < \bar{\tau}(\mu_j)$, the expected accumulated evidence at the epoch boundary is below the SPRT threshold: the evidence accumulation rate is insufficient for reliable detection within the epoch.

The maximum undetected contraction per epoch is $|\mu_j| \cdot \tau_{\text{adapt}}$. For a scorpion to evade detection within an epoch, it must have $\tau_{\text{adapt}} < \bar{\tau}(\mu_j)$. The per-step undetected contraction is bounded by $|\mu_j|$, and the scorpion can sustain this for at most τ_{adapt} steps before changing strategy. At the design point $|\mu_j| = \epsilon$, the per-step contraction is at most ϵ. The time-averaged undetected contraction rate is therefore at most $|\mu_j|$, achievable only by a scorpion that never needs to pause for strategy adaptation. ■